D **3.1-14.** Consider the following problem, where the values of c_1 and c_2 have not yet been ascertained.

$$\text{Maximize} \quad Z = c_1x_1 + c_2x_2,$$

subject to

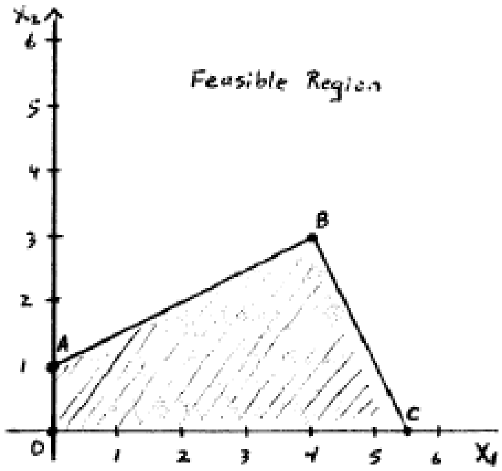
$$\begin{aligned} 2x_1 + x_2 &\leq 11 \\ -x_1 + 2x_2 &\leq 2 \end{aligned}$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$

Use graphical analysis to determine the optimal solution(s) for (x_1, x_2) for the various possible values of c_1 and c_2. (*Hint:* Separate the cases where $c_2 = 0$, $c_2 > 0$, and $c_2 < 0$. For the latter two cases, focus on the ratio of c_1 to c_2.)

3.1-14.



<u>Case 1:</u> $c_2 = 0$ (vertical objective line)

If $c_1 > 0$, the objective value increases as x_1 increases, so $x^* = (\frac{11}{2}, 0)$, point C.

If $c_1 < 0$, the opposite is true so that all the points on the line from $(0,0)$ to $(0,1)$, line $\overline{OA}$, are optimal.

If $c_1 = 0$, the objective function is $0x_1 + 0x_2 = 0$ and every feasible point is optimal.

<u>Case 2:</u> $c_2 > 0$ (objective line with slope $-\frac{c_1}{c_2}$)

If $-\frac{c_1}{c_2} > \frac{1}{2}$ $x^* = (0,1)$, point A.

If $-\frac{c_1}{c_2} < -2$ $x^* = (\frac{11}{2}, 0)$, point C.

If $\frac{1}{2} > -\frac{c_1}{c_2} > -2$ $x^* = (4,3)$, point B.

If $-\frac{c_1}{c_2} = \frac{1}{2}$, any point on the line $\overline{AB}$ is optimal. Similarly, if $-\frac{c_1}{c_2} = -2$, any point on the line $\overline{BC}$ is optimal.

<u>Case 3:</u> $c_2 < 0$ (objective line with slope $-\frac{c_1}{c_2}$, objective value increases as the line is shifted down)

If $-\frac{c_1}{c_2} > 0$, i.e., $c_1 > 0$, $x^* = (\frac{11}{2}, 0)$, point C.

If $-\frac{c_1}{c_2} < 0$, i.e., $c_1 < 0$, $x^* = (0,0)$, point O.

If $-\frac{c_1}{c_2} = 0$, i.e., $c_1 = 0$, x^* is any point on the line $\overline{OC}$.

3.5-6. Maureen Laird is the chief financial officer for the Alva Electric Co., a major public utility in the midwest. The company has scheduled the construction of new hydroelectric plants 5, 10, and 20 years from now to meet the needs of the growing population in the region served by the company. To cover at least the construction costs, Maureen needs to invest some of the company's money now to meet these future cash-flow needs. Maureen may purchase only three kinds of financial assets, each of which costs $1 million per unit. Fractional units may be purchased. The assets produce income 5, 10, and 20 years from now, and that income is needed to cover at least minimum cash-flow requirements in those years. (Any excess income above the minimum requirement for each time period will be used to increase dividend payments to shareholders rather than saving it to help meet the minimum cash-flow requirement in the next time period.) The following table shows both the amount of income generated by each unit of each asset and the minimum amount of income needed for each of the future time periods when a new hydroelectric plant will be constructed.

	Income per Unit of Asset			
Year	**Asset 1**	**Asset 2**	**Asset 3**	**Minimum Cash Flow Required**
5	$2 million	$1 million	$0.5 million	$400 million
10	$0.5 million	$0.5 million	$1 million	$100 million
20	0	$1.5 million	$2 million	$300 million

Maureen wishes to determine the mix of investments in these assets that will cover the cash-flow requirements while minimizing the total amount invested.

(a) Formulate a linear programming model for this problem.

(b) Display the model on a spreadsheet.

(c) Use the spreadsheet to check the possibility of purchasing 100 units of Asset 1, 100 units of Asset 2, and 200 units of Asset 3. How much cash flow would this mix of investments generate 5, 10, and 20 years from now? What would be the total amount invested?

(d) Take a few minutes to use a trial-and-error approach with the spreadsheet to develop your best guess for the optimal solution. What is the total amount invested for your solution?

C (e) Use the Excel Solver to solve the model by the simplex method.

C **3.6-6.** Reconsider Prob. 3.5-6. Use Gurobi to formulate and solve the model for this problem.

3.5-6.

(a) m inimize $C = x_1 + x_2 + x_3$

subject to $2x_1 + x_2 + 0.5x_3 \geq 400$

$0.5x_1 + 0.5x_2 + x_3 \geq 100$

$1.5x_2 + 2x_3 \geq 300$

and $x_1, x_2, x_3 \geq 0$

(b) - (e)

	Benefit Contribution Per Unit of Each Asset					Minimum Cash Flow
Year	Asset 1	Asset 2	Asset 3	Totals		Required
5	2	1	0.5	400	≥	400
10	0.5	0.5	1	150	≥	100
20	0	1.5	2	300	≥	300
Unit Cost	1	1	1	**$ 300**		
Solution	**100**	**200**	**0**			

(c) $(x_1, x_2, x_3) = (100, 100, 200)$ is a feasible solution. This would generate \$400 million in 5 years, \$300 m illion in 10 years, and \$550 million in 20 years. The total investm ent will be \$400 million.

(d) Answers will vary.

```
from gurobipy import Model, GRB
# Create a new model
model = Model("LP_Minimization")

# Create variables (non-negative by default with lb=0)
x1 = model.addVar(lb=0, name="x1")
x2 = model.addVar(lb=0, name="x2")
x3 = model.addVar(lb=0, name="x3")

# Set objective: Minimize C = x1 + x2 + x3
model.setObjective(x1 + x2 + x3, GRB.MINIMIZE)

# Add constraints
model.addConstr(2*x1 + x2 + 0.5*x3 >= 400, "c1")
model.addConstr(0.5*x1 + 0.5*x2 + x3 >= 100, "c2")
model.addConstr(1.5*x2 + 2*x3 >= 300, "c3")

# Optimize model
model.optimize()
# Print results
if model.status == GRB.OPTIMAL:
  print(f"\nOptimal Solution Found:")
  print(f"x1 = {x1.X:.4f}")
  print(f"x2 = {x2.X:.4f}")
  print(f"x3 = {x3.X:.4f}")
  print(f"Minimum C = {model.ObjVal:.4f}")
    print(f"\nShadow Prices:") # print shadow prices
  for constr in model.getConstrs():
    print(f"  {constr.ConstrName}: {constr.Pi:.4f}")
else:    print("No optimal solution found.")
```

■ CASE 3.3 STAFFING A CALL CENTER[1]

California Children's Hospital has been receiving numerous customer complaints because of its confusing, decentralized appointment and registration process. When customers want to make appointments or register child patients, they must contact the clinic or department they plan to visit. Several problems exist with this current strategy. Parents do not always know the most appropriate clinic or department they must visit to address their children's ailments. They therefore spend a significant amount of time on the phone being transferred from clinic to clinic until they reach the most appropriate clinic for their needs. The hospital also does not publish the phone numbers of all clinic and departments, and parents must therefore invest a large amount of time in detective work to track down the correct phone number. Finally, the various clinics and departments do not communicate with each other. For example, when a doctor schedules a referral with a colleague located in another department or clinic, that department or clinic almost never receives word of the referral. The parent must contact the correct department or clinic and provide the needed referral information.

In efforts to reengineer and improve its appointment and registration process, the children's hospital has decided to centralize the process by establishing one call center devoted exclusively to appointments and registration. The hospital is currently in the middle of the planning stages for the call center. Jin Chin, the hospital manager, plans to operate the call center from 7 A.M. to 9 P.M. during the weekdays.

Several months ago, the hospital hired an ambitious management consulting firm, Creative Chaos Consultants, to forecast the number of calls the call center would receive each hour of the day. Since all appointment and registration-related calls would be received by the call center, the consultants decided that they could forecast the calls at the call center by totaling the number of appointment and registration-related calls received by all clinics and departments. The team members visited all the clinics and departments, where they diligently recorded every call relating to appointments and registration. They then totaled these calls and altered the totals to account for calls missed during data collection. They also altered totals to account for repeat calls that occurred when the same parent called the hospital many times because of the confusion surrounding the decentralized process. Creative Chaos Consultants determined the average number of calls the call center should expect during each hour of a weekday. The following table provides the forecasts.

Work Shift	Average Number of Calls
7 A.M.–9 A.M.	40 calls per hour
9 A.M.–11 A.M.	85 calls per hour
11 A.M.–1 P.M.	70 calls per hour
1 P.M.–3 P.M.	95 calls per hour
3 P.M.–5 P.M.	80 calls per hour
5 P.M.–7 P.M.	35 calls per hour
7 P.M.–9 P.M.	10 calls per hour

After the consultants submitted these forecasts, Jin became interested in the percentage of calls from Spanish speakers since the hospital services many Hispanic patients. Jin knows that he has to hire some operators who speak Spanish to handle these calls. The consultants performed further data collection and determined that on average, 20 percent of the calls were from Spanish speakers.

Given these call forecasts, Jin must now decide how to staff the call center during each 2 hour shift of a weekday. During the forecasting project, Creative Chaos Consultants closely observed the operators working at the individual clinics and departments and determined the number of calls operators process per hour. The consultants informed Jin that an operator is able to process an average of six calls per

For the following analysis consider only the labor cost for the time employees spend answering phones. The cost for paperwork time is charged to other cost centers.

(a) How many Spanish-speaking operators and how many English-speaking operators does the hospital need to staff the call center during each 2-hour shift of the day in order to answer all calls? Please provide an integer number since half a human operator makes no sense.

(b) Jin needs to determine how many full-time employees who speak Spanish, full-time employees who speak English, and part-time employees he should hire to begin on each shift. Creative Chaos Consultants advise him that linear programming can be used to do this in such a way as to minimize operating costs while answering all calls. Formulate a linear programming model of this problem.

hour. Jin also knows that he has both full-time and part-time workers available to staff the call center. A full-time employee works 8 hours per day, but because of paperwork that must also be completed, the employee spends only 4 hours per day on the phone. To balance the schedule, the employee alternates the 2-hour shifts between answering phones and completing paperwork. Full-time employees can start their day either by answering phones or by completing paperwork on the first shift. The full-time employees speak either Spanish or English, but none of them are bilingual. Both Spanish-speaking and English-speaking employees are paid \$20 per hour for work before 5 P.M. and \$24 per hour for work after 5 P.M. The full-time employees can begin work at the beginning of the 7 A.M. to 9 A.M. shift, 9 A.M. to 11 A.M. shift, 11 A.M. to 1 P.M. shift, or 1 P.M. to 3 P.M. shift. The part-time employees work for 4 hours, only answer calls, and only speak English. They can start work at the beginning of the 3 P.M. to 5 P.M. shift or the 5 P.M. to 7 P.M. shift, and like the full-time employees, they are paid \$20 per hour for work before 5 P.M. and \$24 per hour for work after 5 P.M.

(c) Obtain an optimal solution for the linear programming model formulated in part (*b*) to guide Jin's decision.

(d) Because many full-time workers do not want to work late into the evening, Jin can find only one qualified English-speaking operator willing to begin work at 1 P.M. Given this new constraint, how many full-time English-speaking operators, full-time Spanish-speaking operators, and part-time operators should Jin hire for each shift to minimize operating costs while answering all calls?

(e) Jin now has decided to investigate the option of hiring bilingual operators instead of monolingual operators. If all the operators are bilingual, how many operators should be working during each 2-hour shift to answer all phone calls? As in part (*a*), please provide an integer answer.

(f) If all employees are bilingual, how many full-time and part-time employees should Jin hire to begin on each shift to minimize operating costs while answering all calls? As in part (*b*), formulate a linear programming model to guide Jin's decision.

(g) What is the maximum percentage increase in the hourly wage rate that Jin can pay bilingual employees over monolingual employees without increasing the total operating costs?

(h) What other features of the call center should Jin explore to improve service or minimize operating costs?

3-3 a) The number of operators that the hospital needs to staff the call center during each two-hour shift can be found in the following table:

	A	B	C	D	E	F
1	work	average number of	number of calls from	number of calls from	number of operators	number of operators
2	shift	calls per hour	English speakers	Spanish speakers	speaking English	speaking Spanish
3	7am to 9am	40	32	8	6	2
4	9am to 11am	85	68	17	12	3
5	11am to 1pm	70	56	14	10	3
6	1pm to 3pm	95	76	19	13	4
7	3pm to 5pm	80	64	16	11	3
8	5pm to 7pm	35	28	7	5	2
9	7pm to 9pm	10	8	2	2	1

For example, the average number of phone calls per hour during the shift from 7am to 9am equals 40. Since, on average, 80% of all phone calls are from English speakers, there is an average number of 32 phone calls per hour from English speakers during that shift. Since one operator takes, on average, 6 phone calls per hour, the hospital needs $32/6 = 5.333$ English-speaking operators during that shift. The hospital cannot employ fractions of an operator and so needs 6 English-speaking operators for the shift from 7am to 9am.

b) The problems of determining how many Spanish-speaking operators and English-speaking operators Lenny needs to hire to begin each shift are independent. Therefore we can formulate two smaller linear programming models instead of one large model. We are going to have one model for the scheduling of the Spanish-speaking operators and another one for the scheduling of the English-speaking operators.

Lenny wants to minimize the operating costs while answering all phone calls. For the given scheduling problem we make the assumption that the only operating costs are the wages of the employees for the hours that they answer phone calls. The wages for the hours during which they perform paperwork are paid by other cost centers. Moreover, it does not matter for the callers whether an operator starts his or her work day with phone calls or with paperwork. For example, we do not need to distinguish between operators who start their day answering phone calls at 9am and operators who start their day with paperwork at 7am, because both groups of operators will be answering phone calls at the same time. And only this time matters for the analysis of Lenny's problem.

We define the decision variables according to the time when the employees have their first shift of answering phone calls. For the scheduling problem of the English-speaking operators we have 7 decision variables. First, we have 5 decision variables for full-time employees.

The number of operators having their first shift on the phone from 7am to 9am.
The number of operators having their first shift on the phone from 9am to 11am.
The number of operators having their first shift on the phone from 11am to 1pm.
The number of operators having their first shift on the phone from 1pm to 3pm.
The number of operators having their first shift on the phone from 3pm to 5pm.

In addition, we define 2 decision variables for part-time employees.

The number of part-time operators having their first shift from 3pm to 5pm.
The number of part-time operators having their first shift from 5pm to 7pm.

The unit cost coefficients in the objective function are the wages operators earn while they answer phone calls. All operators who have their first shift on the phone from 7am to 9am, 9am to 11am, or 11am to 1pm finish their work on the phone before 5pm. They earn 4*\$10 = \$40 during their time answering phone calls. All operators who have their first shift on the phone from 1pm to 3pm or 3pm to 5pm have one shift on the phone before 5pm and another one after 5pm. They earn 2*\$10+2*\$12 = \$44 during their time answering phone calls. The second group of part-time operators, those having their first shift from 5pm to 7pm, earn 4*\$12 = \$48 during their time answering phone calls.

There are 7 constraints, one for each two-hour shift during which phone calls need to be answered. The right-hand sides for these constraints are the number of operators needed to ensure that all phone calls get answered in a timely manner. On the left-hand side we determine the number of operators on the phone during any given shift. For example, during the 11am to 1pm shift the total number of operators answering phone calls equals the sum of the number of operators who started answering calls at 7am and are currently in their second shift of the day and the number of operators who started answering calls at 11am.

The following spreadsheet describes the entire problem formulation for the English-speaking employees:

	A	B	C	D	E	F	G	H	I	J	K
1											
2	Shifts of	Number of operators whose first shift of answering phone calls in English is from									Required number
3	phone operators	7am to 9am	9am to 11am	11am to 1 pm	1pm to 3pm	3pm to 5pm	3pm to 5pm (P)	5pm to 7pm (P)	Totals		of operators
4	7am to 9am	1	0	0	0	0	0	0	6	>=	6
5	9am to 11am	0	1	0	0	0	0	0	13	>=	12
6	11am to 1 pm	1	0	1	0	0	0	0	10	>=	10
7	1pm to 3pm	0	1	0	1	0	0	0	13	>=	13
8	3pm to 5pm	0	0	1	0	1	1	0	11	>=	11
9	5pm to 7pm	0	0	0	1	0	1	1	5	>=	5
10	7pm to 9pm	0	0	0	0	1	0	1	2	>=	2
11	Unit cost	40	40	40	44	44	44	48	**1228**	=	Total cost
12	Solution	**6**	**13**	**4**	**0**	**2**	**5**	**0**			

And for Spanish speakers:

	A	B	C	D	E	F	G	H	I	J
1										
2		Shifts of	Number of operators whose first shift of answering phone calls in Spanish is from							Required number
3		phone operators	7am to 9am	9am to 11am	11am to 1 pm	1pm to 3pm	3pm to 5pm	Totals		of operators
4		7am to 9am	1	0	0	0	0	2	>=	2
5		9am to 11am	0	1	0	0	0	3	>=	3
6		11am to 1 pm	1	0	1	0	0	4	>=	3
7		1pm to 3pm	0	1	0	1	0	5	>=	4
8		3pm to 5pm	0	0	1	0	1	3	>=	3
9		5pm to 7pm	0	0	0	1	0	2	>=	2
10		7pm to 9pm	0	0	0	0	1	1	>=	1
11		Unit cost	40	40	40	44	44	**412**	=	Total cost
12		Solution	**2**	**3**	**2**	**2**	**1**			

c) Lenny should hire 25 full-time English-speaking operators. Of these operators, 6 have their first phone shift from 7am to 9am, 13 from 9am to 11am, 4 from 11am to 1pm, and 2 from 3pm to 5pm. Lenny should also hire 5 part-time operators who start their work at 3pm. In addition, Lenny should hire 10 Spanish-speaking operators. Of these operators, 2 have their first shift on the phone from 7am to 9am, 3 from 9am to 11am, 2 from 11am to 1pm and 1pm to 3pm, and 1 from 3pm to 5pm. The total (wage) cost of running the calling center equals $1640 per day.

d) The restriction that Lenny can find only one English-speaking operator who wants to start work at 1pm affects only the linear programming model for English-speaking operators. This restriction does not put a bound on the number of operators who start their first phone shift at 1pm because those operators can start work at 11am with paperwork. However, this restriction does put an upper bound on the number of operators having their first phone shift from 3pm to 5pm. The new worksheet appears as follows.

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2		Shifts of	Number of operators whose first shift of answering phone calls in English is from									Required number
3		phone operators	7 am to 9am	9am to 11am	11am to 1 pm	1pm to 3pm	3pm to 5pm	3pm to 5pm (P)	5 pm to 7pm (P)	Totals		of operators
4		7am to 9am	1	0	0	0	0	0	0	6	>=	6
5		9am to 11am	0	1	0	0	0	0	0	13	>=	12
6		11am to 1 pm	1	0	1	0	0	0	0	12	>=	10
7		1pm to 3pm	0	1	0	1	0	0	0	13	>=	13
8		3pm to 5pm	0	0	1	0	1	1	0	11	>=	11
9		5pm to 7pm	0	0	0	1	0	1	1	5	>=	5
10		7pm to 9pm	0	0	0	0	1	0	1	2	>=	2
11		Unit cost	40	40	40	44	44	44	48	1268	=	Total cost
12		Solution	6	13	6	0	1	4	1			
13		Upper bounds					1					

Lenny should hire 26 full-time English-speaking operators. Of these operators, 6 have their first phone shift from 7am to 9am, 13 from 9am to 11am, 6 from 11am to 1pm, and 1 from 3pm to 5pm. Lenny should also hire 4 part-time operators who start their work at 3pm and 1 part-time operator starting work at 5pm. The hiring of Spanish-speaking operators is unaffected. The new total (wage) costs equal $1680 per day.

e) For each hour, we need to divide the average number of calls per hour by the average processing speed, which is 6 calls per hour. The number of bilingual operators that the hospital needs to staff the call center during each two-hour shift can be found in the following table:

	A	B	C
1	work	average number of	number of operators
2	shift	calls per hour	speaking English
3	7am to 9am	40	7
4	9am to 11am	85	15
5	11am to 1pm	70	12
6	1pm to 3pm	95	16
7	3pm to 5pm	80	14
8	5pm to 7pm	35	6
9	7pm to 9pm	10	2

f) The linear programming model for Lenny's scheduling problem can be found in the same way as before, only that now all operators are bilingual.

	A	B	C	D	E	F	G	H	I	J	K
1											
2	Shifts of	Number of operators whose first shift of answering phone calls in both languages is from									Required number
3	phone operators	7am to 9am	9am to 11am	11am to 1pm	1pm to 3pm	3pm to 5pm	3pm to 5pm (P)	5pm to 7pm (P)	Totals		of operators
4	7am to 9am	1	0	0	0	0	0	0	7	>=	7
5	9 am to 11am	0	1	0	0	0	0	0	16	>=	15
6	11am to 1 pm	1	0	1	0	0	0	0	13	>=	12
7	1pm to 3pm	0	1	0	1	0	0	0	16	>=	16
8	3pm to 5pm	0	0	1	0	1	1	0	14	>=	14
9	5pm to 7pm	0	0	0	1	0	1	1	6	>=	6
10	7pm to 9pm	0	0	0	0	1	0	1	2	>=	2
11	Unit cost	40	40	40	44	44	44	48	**1512**	=	Total cost
12	Solution	**7**	**16**	**6**	**0**	**2**	**6**	**0**			

(The formulas and the solver dialogue box are identical to those in part (b).)

Lenny should hire 31 full-time bilingual operators. Of these operators, 7 have their first phone shift from 7am to 9am, 16 from 9am to 11am, 6 from 11am to 1pm, and 2 from 3pm to 5pm. Lenny should also hire 6 part-time operators who start their work at 3pm. The total (wage) cost of running the calling center equals $1512 per day.

g) The total cost of part (f) is $1512 per day; the total cost of part (b) is $1640. Lenny could pay an additional $1640-$1512 = $128 in total wages to the bilingual operators without increasing the total operating cost beyond those for the scenario with only monolingual operators. The increase of $128 represents a percentage increase of 128/1512 = 8.466%.

h) Creative Chaos Consultants has made the assumption that the number of phone calls is independent of the day of the week. But maybe the number of phone calls is very different on a Monday than it is on a Friday. So instead of using the same number of average phone calls for every day of the week, it might be more appropriate to determine whether the day of the week affects the demand for phone operators. As a result Lenny might need to hire more part-time employees for some days with an increased calling volume.

Similarly, Lenny might want to take a closer look at the length of the shifts he has scheduled. Using shorter shift periods would allow him to "fine tune" his calling centers and make it more responsive to demand fluctuations.

Lenny should investigate why operators are able to answer only 6 phone calls per hour. Maybe additional training of the operators could enable them to answer phone calls quicker and so increase the number of phone calls they are able to answer in an hour.

Finally, Lenny should investigate whether it is possible to have employees switching back and forth between paperwork and answering phone calls. During slow times phone operators could do some paperwork while they are sitting next to a phone, while in times of sudden large call volumes employees who are scheduled to do paperwork could quickly switch to answering phone calls.

Lenny might also want to think about the installation of an automated answering system that gives callers a menu of selections. Depending upon the caller's selection, the call is routed to an operator who specializes in answering questions about that selection.

5.1-8. Each of the following statements is true under most circumstances, but not always. In each case, indicate when the statement will not be true and why.

(a) The best CPF solution is an optimal solution.

(b) An optimal solution is a CPF solution.

(c) A CPF solution is the only optimal solution if none of its adjacent CPF solutions are better (as measured by the value of the objective function).

5.1-9. Consider the original form (before augmenting) of a linear programming problem with *n* decision variables (each with a nonnegativity constraint) and *m* functional constraints. Label each of the following statements as true or false, and then justify your answer with specific references (including page citations) to material in the chapter.

(a) If a feasible solution is optimal, it must be a CPF solution.

(b) The number of CPF solutions is at least $\binom{m+n}{n} = \frac{(m+n)!}{m!n!}$.

(c) If a CPF solution has adjacent CPF solutions that are better (as measured by *Z*), then one of these adjacent CPF solutions must be an optimal solution.

5.1-10. Label each of the following statements about linear programming problems as true or false, and then justify your answer.

(a) If a feasible solution is optimal but not a CPF solution, then infinitely many optimal solutions exist.

(b) If the value of the objective function is equal at two different feasible points $\mathbf{x}^*$ and $\mathbf{x}^{**}$, then all points on the line segment connecting $\mathbf{x}^*$ and $\mathbf{x}^{**}$ are feasible and *Z* has the same value at all those points.

(c) If the problem has *n* variables (before augmenting), then the simultaneous solution of any set of *n* constraint boundary equations is a CPF solution.

5.1-11. Consider the augmented form of linear programming problems that have feasible solutions and a bounded feasible region. Label each of the following statements as true or false, and then justify your answer by referring to specific statements (with page citations) in the chapter.

(a) There must be at least one optimal solution.

(b) An optimal solution must be a BF solution.

5.1-8.

(a) If the feasible region is unbounded, then there may be no optimal solution.

(b) There may be multiple optimal solutions, in which case the weighted average of any optimal CPF solutions is optimal, too.

(c) An adjacent CPF solution may have an equal objective function value, then all the points that lie on the line segment between these two corner points are optimal.

5.1-9.

(a) FALSE. (p.5-10) Property 1: (a) If there is exactly one optimal solution, then it must be a CPF solution. (b) If there are multiple optimal solutions, then at least two of them must be adjacent CPF solutions. An optimal solution that is not a CPF solution can be obtained by taking a convex combination of two optimal CPF solutions.

(b) FALSE. (p.5-12) The number of CPF solutions is at most $\begin{pmatrix} m+n \\ n \end{pmatrix} = \frac{(m+n)!}{m!n!}$.

(c) FALSE. (p.5-13) The adjacent CPF solution that has a better objective function value than the initial CPF solution may be adjacent to another CPF solution that has an even better objective function value.

5.1-10.

(a) TRUE. By Property 1(a), there must be multiple solutions, since this optimal solution is not a CPF solution. But then, there must be infinitely many optimal solutions, namely any convex combination of optimal solutions.

(b) TRUE. Any point x on the line segment connecting x^* and x^{**} can be expressed as $x = \alpha x^* + (1-\alpha)x^{**}$ with $\alpha \in [0,1]$. Both x^* and x^{**} have the optimal objective value Z^*. The objective function value at x is

$$Z = c^T(\alpha x^* + (1-\alpha)x^{**}) = \alpha Z^* + (1-\alpha)Z^* = Z^*,$$

so x is optimal. Since the feasible region is convex, any such point is feasible.

(c) FALSE. The simultaneous solution of any set of n constraint boundary equations may be infeasible or may not even exist.

5.1-11.

(a) TRUE. If there are no optimal solutions, then either the problem is infeasible or the objective value is unbounded (Chapter 3). The former is not the case by assumption of the problem. Also by assumption again, the feasible region is bounded, so the objective value is bounded, so the latter cannot be the case. Hence, there must be at least one optimal solution.

(b) FALSE. If a solution is optimal, it need not be a BF solution. A convex combination of two optimal BF solutions is optimal even though it is not a BF solution. This follows from Property 1, since BF solutions are CPF solutions.

(c) TRUE. Since BF solutions correspond to CPF solutions, this follows directly from Property 2.